

Prediction of Power Consumption in Smart Grid: A Reliable Path to a Smart City Based on Various Machine Learning Models

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Abstract: The smart grid is a technology that was created today to address the problem of maintaining significant quantities of energy consumption with the help of emerging nations. Smart cities, like the smart grid, have an energy infrastructure that is arguably the single most important feature in any city. Smart cities depend on a smart grid to ensure resilient delivery of energy to supply their many functions, including those responsible for public safety and the public. It consists of smart distribution boards and circuit breakers integrated with home control and demand response, where load control switches and smart appliances are located. The smart grid vision is supported by the use of information and communication technology. In this article, the capabilities and technologies of the smart grid are compared across several machine learning and deep learning technologies. The deep learning (ANN) methodology was used to find the best approaches for forecasting grid stability. Various supervised machine learning classifiers are utilized to estimate accuracy, AUC-ROC, and other metrics. According to the investigation, deep learning-based estimations were actually more accurate than supervised learning-based ones. As a result, accurate power consumption estimation makes sure the entire chain runs smoothly.

Keywords—Machine learning, Smart grid, Smart City AI, Deep learning, Stability prediction.

I. INTRODUCTION

The construction and maintenance of the smart grid will include sourcing materials from both industrialized and developing nations. The old grid consisted of a few production centers that delivered energy to consumers and had unidirectional flows between them before the advent of renewable energy. However, when renewable energy became more prevalent, a hybrid of the two known as "prosumers," who both use and gain electricity, emerged. Alternative sources are associated with more "prosumers" and more challenging

production, transportation, and utilization. To improve performance and the quality of the energy transmission, the smart grid (SG) may make adjustments on its own. The SG was also excellent at foreseeing issues and disruptions. The SG will then consider a more dependable power source, a reduced cost per kilowatt, and rapid recovery. For these numerous enterprises to receive the appropriate amount of electricity, the smart grid is crucial. Smart grids consist of mainly three components, which are presented below in Fig.1.

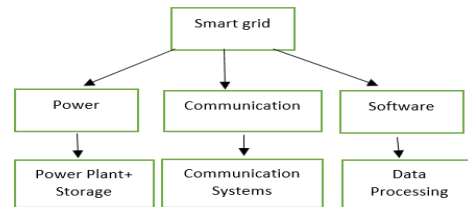


Fig. 1. Smart grid's components

In the smart grid process, end-user data is gathered, compared to current grid data, and used to supply data, estimate a price for this electricity, and communicate this information to users. One of the most crucial aspects of such a system is that it gradually assesses grid stability because the interaction is time dependent. End-users can calculate how much energy consumers use by collecting the charge rate. End-use consumers can play a significant role in smart grid activity by adjusting their power usage in reaction to variations in power costs over time [1-3].

In the present day, most cities have converted to "smart cities" based on updated technology applied in every sector to improve the lifestyle. Day by day, natural resources are diminishing, but on the other hand, researchers are trying to find an efficient way to implement the available sources. Machine learning is applied to estimating demand and what amount of power is required in the smart city, which is a new concept [4,5]. Different supervised ML methods were used in this study to compare their stability performance analysis [6]. Distributed smart grid monitoring just needs frequency components that are readily available elsewhere in the network, and compares different deep learning methods with

poor performance when considerable simplification is taken into account [7,8].

Other common Deep Learning algorithms, such as Gated Recurrent Units (GRU), conventional LSTM [9], and Recurrent Neural Networks are compared to the results produced. AI has been used in a number of areas that are important for the reliability and consistency of a smart grid [10].

To help readers understand the complexity, possible advantages, and difficulties associated with this fascinating topic, the essay first introduces the idea of the "smart grid" and its different expressions. Second, a quick overview of current projects, advances, technologies, and research is provided in the article mentioned in the manuscripts listed under the application of smart grid systems in smart cities. Then, to determine the grid system's stability position, we are evaluating several machines and deep learning methodologies. The essay then finished with a quick explanation of some potential future scope and the various model performances of the utilized dataset.

II. PROPOSED METHODOLOGY

In this section, we discussed about our proposed methodology

A. Dataset used

Simulated Electrical Grid Stability Data, a machine learning repository at UC Irvine, is where it was gathered. Data set for the electrical grid with various attributes to test the system's stability and contains results from simulations of grid stability for a reference 4-node star network. Nave Bayes, Random forest, and decision trees are used in the study to classify the system stability for various sets of input values. We will investigate the system's response to its stability, which is dependent on 10,000 observations, 13 attributes, and one class attribute (stab), which is the node shown in Fig. 2.

As the reference grid is symmetric, the dataset can be augmented three (three factorial) times or six times, representing a permutation of the three consumers occupying three consumer nodes. The augmented version then has 60,000 observations. It also contains 12 primary predictive features and two dependent variables. This talk will only discuss how to categorize the stability of the electrical grid into two categories:

1. Stable 2. Unstable

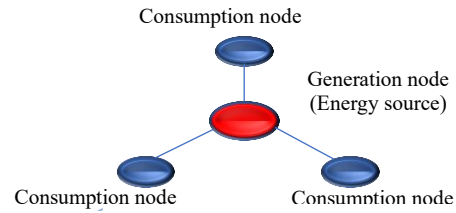


Fig. 2. Reference 4 node star network

B. Data Pre-processing

14 predictive attributes altogether, 1 target attribute (2classes)

- $\tau[x]$: Participant's reaction time (real from the range $[0.5,10]$ s).
- $p[x]$: Nominal power produced/consumed (negative) (real).
- $g[x]$: Gamma coefficient (real from $[0.05,1]$ s-1) proportional to price elasticity.
- stab: The largest real component of the root of the characteristic equation (if positive, the system is linearly unstable) (real)
- stabf: The system's stability label (stable/unstable)

Data encoding is performed during data pre-processing, which converts non-numeric values to numeric values before being fed to Machine learning because most of the dataset contains both numerical and category data. As there is a direct relationship between 'stab' and 'stabf' ('stabf' = 'stable' if 'stab' ≤ 0 , 'unstable' otherwise), 'stab' will be dropped and 'stabf' will remain as the sole dependent variable. In the grid data, the stable grid condition is indicated as "1," whereas the unstable grid condition is labeled as "0." There are no missing numbers because the content of the dataset is drawn from simulated exercises. After that, training and testing datasets are created from the pre-processed data. The ML algorithms are trained using the training dataset, and their performance is then tested using a fresh batch of data. The dataset is split into two parts: 80% is applied to train the ML algorithms, and the remaining 20% is used to evaluate the effectiveness of the control ML algorithms.

C. Implementation of the ML Model

For predicting the stability of the smart grid, the following ML algorithms have been implemented:

1. Decision Tree (DT)
2. K-Nearest Neighbor (KNN)
3. Support Vector Machine (SVM)
4. Logistic Regression (LR)
5. Ensemble Soft Voting (ESV) method
6. Random Forest (RF)
7. Naïve Bayes (NB)
8. XGBoost
9. GBM
10. Artificial Neural Network (ANN)

The proposed workflow has been illustrated in fig. 3

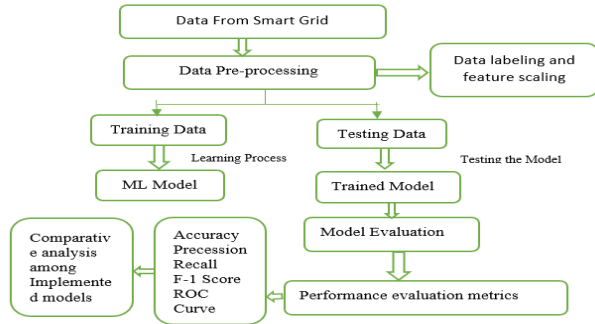


Fig. 3. System Architecture

The neural network maps the evaluation with standard measurements to the proper result unit after executing specific numerical calculations. A hidden layer holds elements which are presented as information, and stowed away layers are positioned between the information and output layers in the neural network. The hidden layer's function is to multiply the input layer's weights before passing the resulting values to the activation functions. The final layer, known as the outcome layer, is where the forecasts are carried out. Fig.4 depicts the neural network's entire design.

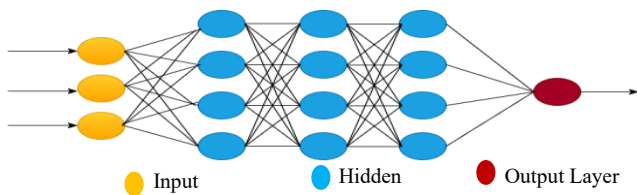


Fig. 4. Neural network architecture

III. RESULT AND DISCUSSION

Based on the following assessment metrics, comparative performance study of the various ML models has been carried out:

1. Accuracy 2. Precision 3. Recall 4. F1-score.

A. Metrics for assessing the effectiveness of the various ML models. The performance metrics that were utilized to assess the various ML models are described below.

1. Confusion Matrices

True Positive (TP)	False Positive (FP)
False Negative (FN)	True Negative (TN)

2. Accuracy: $\frac{TP + TN}{TP + FP + TN + FN}$
3. Precision: $\frac{TP}{TP + FP}$
4. Recall: $\frac{TP}{TP + FN}$
5. F1-score: $\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

AUC-ROC: Area Under Curve- Receiver Operating Characteristics as an evaluation metric for binary classification issues is the AUC-ROC curve. The classifier's capacity to distinguish between the two classes is gauged by the area under the curve (AUC). The False Positive rate is shown by the X-axis of the AUC-ROC curve, while the True Positive rate is shown by the y-axis.

$$TPR = \frac{TP}{TP + FN} = \text{Recall}$$

The False Positive Rate (FPR) helps in understanding what proportion of the negative class got correctly classified by the classifier.

$$\begin{aligned} FPR &= \frac{FP}{FP + TN} \\ &= 1 - \frac{TN}{FP + TN} \\ &= 1 - \text{Specificity} \end{aligned}$$

AUC near to 1 which is an indication that the model has a good measure of separability.

B. Performance evaluation of the implemented algorithms

After the data insert, we reshuffle the whole data because of our binary output. The output will be biased if the initial portion of the data is all stable and the second portion of the data is all unstable, which is why we shuffle the entire data. Although our data includes two goal columns, the 'staff' column describes the same thing as the first column, which has a positive and negative value which defines the last column as positive shows unstable and negative value means stable. That's why we drop the 'stab' columns.

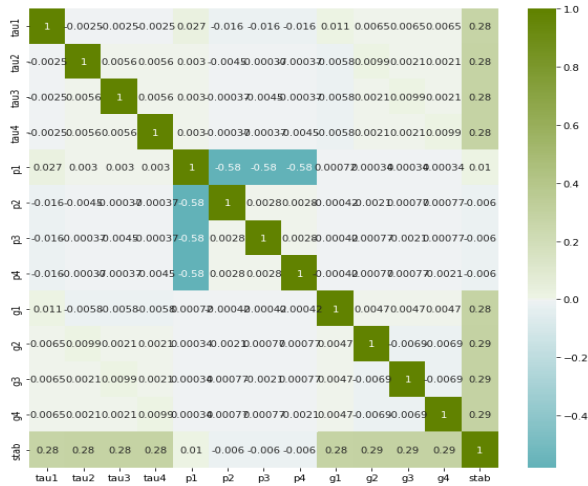


Fig. 5. Correlation matrices

We attempt to eliminate the dependency when examining our dataset's reliance on a certain collection of data. As a result, the accuracy of our deep learning and machine learning model application is exceptional. The correlation matrix assists us in determining which qualities are more relevant than others. From the above output of the correlation matrix (Fig. 5), we can see that it is symmetrical, i.e., the top right and bottom left are identical. All parameters are also found to be positively related. Furthermore, because the dataset has already been shuffled, it will be split 80:20, with the first 48,000 observations going to the training set and the last 12000 going to the testing set. The table shows the evaluation metrics for each of the seven machine learning models. Among them, six are the direct supervised learning classification method, and the last one is the ensemble soft voting method that uses the upper six classifiers' output as a weight value and finally averages this weight to give the output result

TABLE I. CRITERIA FOR CALCULATING THE SEVERAL MODEL'S PERFORMANCE METRICS

Model	TP	FP	FN	TN
SVM	3040	1328	841	6791
DT	2899	1422	734	6945
LR	3043	1278	920	6759
RF	2902	1419	763	6916
KNN	2982	1339	851	6828
NB	3062	1250	945	6734
ESV	3259	1101	419	7221

This value is collected from the confusion metrics. From the Table I, the Decision Tree classifiers has detected 2899 data as true positive and 6945 true negative precisely. It has misclassified only $(1422+734) = 2156$ data. In fig.6 Accuracy = $(2899+6945)/(2899+1422+734+6945) = 0.8203$ and misclassification = $(1422+734)/(2899+1422+734+6945) = 0.1796$.

Actual:0	2899	1422
	Predicted:0	Predicted:1
Actual:1	734	6945
	Predicted:0	Predicted:1

Fig. 6. Confusion matrices for DT Classifier

But this is not the ultimate result. The Ensemble soft voting method uses all six classifiers to identify more correctly fig.7 than any single classifier. In comparison with DT, the confusion metrics of the Voting method.

Actual:0	3259	1101
	Predicted:0	Predicted:1
Actual:1	419	7221
	Predicted:0	Predicted:1

Fig. 7. Confusion matrices for Ensemble Soft voting method

Simply divide the test data into sections and apply seven different machine learning methods, such as decision trees, support vector machines, random forests, logistic regression, and K-Nearest Neighbor. It is accurate to divide the total number of data instances by the number of instances that were correctly categorized. The bar chart in Fig. 8 below displays a comparison of the accuracy of various algorithms fig.8 : The evaluation metrics for each model have been summarized to provide a clear picture.

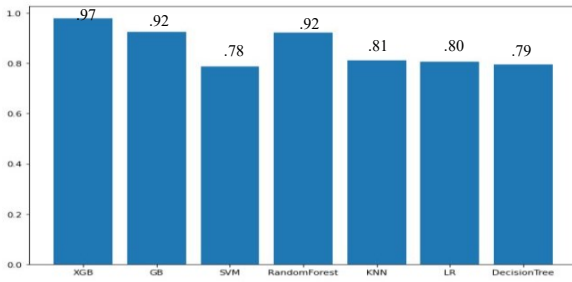


Fig. 8. Machine Learning model Accuracy

Table II provides a performance analysis of various machine learning algorithms based on accuracy, precision, recall, and F-1 Score.

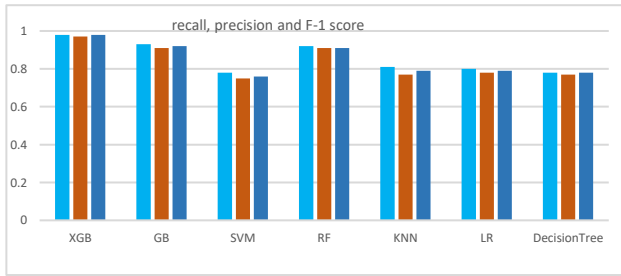


Fig. 9. Performance evaluation based on accuracy, recall, precision and F-1 score

From the Table II, we can be concluded that Ensemble XGB exhibits outperformed all the other model in terms of precision, recall and F-1 score which is 98%, 97%, 86.77 and 98%. But if we compared with single classifiers then Decision Tree fig.9 shows best score with Accuracy is 79.48%, Recall value is 77.4%, Precision is 78% and F-1 score is 78.56%.

Therefore, we conclude that for single classifier of supervised model, DT is more precisely shows result but Soft voting method will be far more accurately predict the output. To enable a clear picture of this binary classification model, AOC ROC has been plotted fig.10 and this has been described below

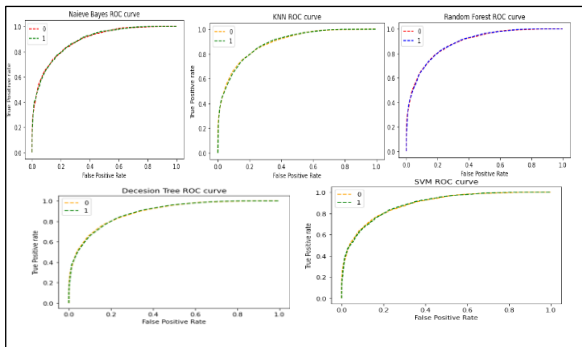


Fig. 10. ROC Curve in ALL Classification

The upper ROC curve is combined with every type of classification. If AUC-ROC is 1, every forecast, whether stable or unstable, will accurately estimate the future value. When we applied each classification, there were some hypermeters that define the optimal performance. We used a grid search to determine which number delivers the highest accuracy when computing this hyper meter.

If we change this hypermeter so much, then the model will be overfitting or underfitting what was never expected in our result.

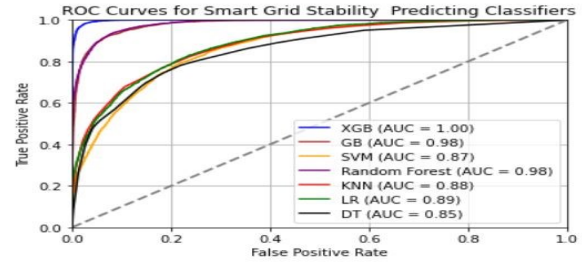


Fig. 11. AUC-ROC Curve in ALL Classification

The area under the ROC curves of all the ML algorithms have been tabulated in Table III

TABLE II. PERFORMANCE COMPARISON OF THE MODEL BASED ON AUC-ROC

Model	AUC
SVM	0.87
DT	0.85
LR	0.89
RF	0.98
KNN	0.88
GB	0.98
XGB	1.0

From the above table III, it can be observed that RF and KNN have the highest AUC-ROC value among all classifiers fig.11. Both yielded a result of 0.98, which is nearly equivalent to one. As a result, they can determine between the 2 classes of stable (1) and unstable (0) more accurately than other classifiers, but the other classifier can also discriminate between the two classes. It can be inferred that the performance of all the classifiers has been quite promising.

The accuracy of supervised learning range below 90%. If we want to improve the accuracy and another parameter, we proposed using deep learning for stability prediction to solve these issues. The simple ANN model is used to predict the output with adam optimizer. A third hidden

layer with dense values of 64, 64, and 32 is added to the Neural model. All of the 100-test values are ignored on each layer, and we discard 20% of the data since we don't want to memorize the data but rather learn to estimate automatically. With 20 epochs and a batch size of 32, the maximum test accuracy is 96.60.

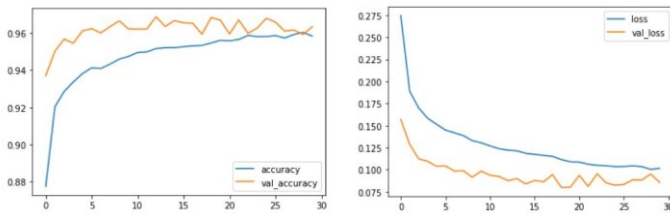


Fig. 12. ANN Accuracy and Loss vs epoch

From Fig. 12, we see that accuracy increases very quickly in the 0 to 25 epoch, and after that, the rate goes slow as the epoch increases, but the validation accuracy does not change abruptly but follows the same direction as actual accuracy. The real loss drop is inversely related to the accuracy decline due to the drastic shift in precision. But at the same time, the validation loss will decrease smoothly as the epoch increases. The confusion matrix value From the confusion matrix, the predicted rate is much higher as compared to the supervised learning method. As a result of the foregoing evaluations, we conclude that using the Artificial Neural Network technique or other deep learning methods to forecast potential values is far superior.

IV. CONCLUSION:

AI and machine learning techniques are extremely useful in smart power distribution systems. So, the correct assumption of power consumption runs this whole chain smoothly and without any disturbances. The grid's stability depends on how precisely it anticipates daily usage. In this work, we try to compare the analysis of seven supervised classifications with one deep neural network, which are SVM, LR, DT, RF, NB, KNN, Soft voting, and ANN, with several evaluation metrics, which are accuracy, recall, precision, F-1 score, and AUC_ROC, whose dataset was obtained from the UCI machine learning repository. Out of the classification methods Ensemble, the soft voting test accuracy is 87.33%, but among the six classifications, DT gives 82.03% accuracy. The accuracy of all six classification methods ranges between 81% and

82%. The accuracy rises to 95.60 % when we switch to the ANN model with three hidden layers. As with all of the preceding analyses, we conclude that using synthetic neural networks or other deep learning techniques for estimation is far more accurate.

V. FUTURE SCOPE

The issue with the existing paintings is that the length of the dataset is small. A special gadget authority set of rules was provided as a portion of predetermination for moving forward with execution. Another important problem is to simulate with one of each type of parameter, generating a dataset, in order to improve stability prediction.

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